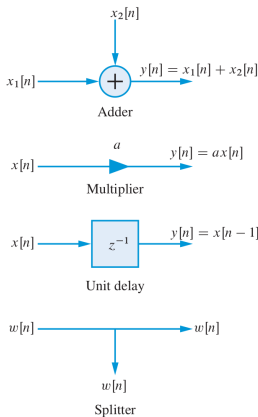
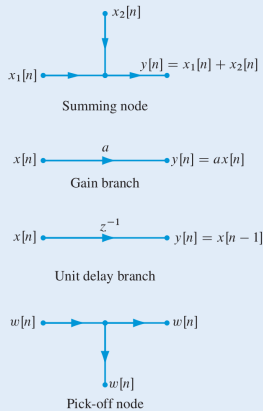


Block diagram & signal flow graph

Block Diagram Elements



Signal Flow Graph Elements



Block Diagram

- Nodes (blocks) → **operations or subsystems**
- Connections → **signals**
- Addition → explicit **summing junctions**
- Branching → explicit **pickoff points**
- Emphasizes **system structure**
- Typical use: **control system modeling**

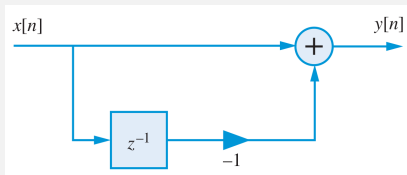
$$x \rightarrow [G(s)] \rightarrow y \quad y = G(s)x$$

Signal Flow Graph

- Nodes → **variables (signals)**
- Directed branches → **gains**
- Addition → **implicitly at nodes**
- Branching → **naturally**
- Emphasizes **algebraic relations**
- Typical use: **calculate gain**

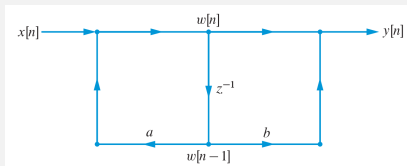
$$x \xrightarrow{G(s)} y \quad y = G(s)x$$

Example block diagram: First difference system



$$y[n] = x[n] - x[n - 1]$$

Example signal graph: General first order system



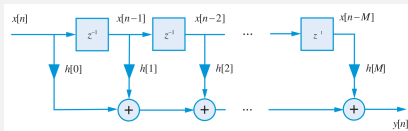
$$\begin{array}{l} w[n] = x[n] + aw[n - 1] , \\ y[n] = w[n] + bw[n - 1] . \end{array} \quad \left| \begin{array}{l} + \\ , \end{array} \right. \begin{array}{l} b \times (\dots) \\ a \times (\dots) \end{array} \left| \begin{array}{l} - \\ - \end{array} \right. \rightarrow \begin{array}{l} (b + a)w[n - 1] \\ (b + a)w[n] \end{array} \quad \begin{array}{l} = y[n] - x[n] , \\ = bx[n] + ay[n] . \end{array} \quad \left. \begin{array}{l} n \rightarrow n - 1 \end{array} \right| -$$

By eliminating $w[n]$ and $w[n - 1]$

$$y[n] = x[n] + bx[n - 1] + ay[n - 1] .$$

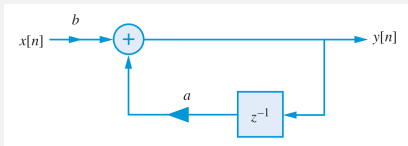
- **Finite (duration) impulse response system (FIR):** the impulse response has a finite number of nonzero samples (finite support)
- **Infinite (duration) impulse response system (IIR):** infinite number of unit delays and arithmetic operations.

Example block diagram: FIR system



$$y[n] = \sum_{k=0}^M h[k]x[n-k]$$

Example block diagram: simple recursive IIR system

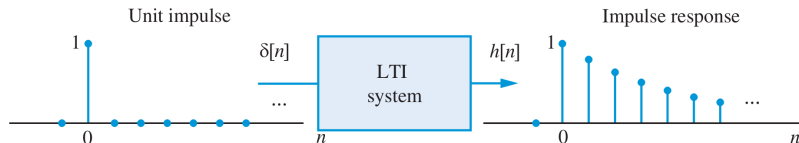


$$y[n] = bx[n] + ay[n-1] .$$

Linear, time-invariant (LTI) system

- **time-invariance** = time independence $\rightarrow$ the only impact of a time shift in the input is a similar shift in the output. This is not the case when the system reacts in time dependent (non-autonomous) way, there is resampling, etc.
- **linearity** means that individual signals which have to be summed at the input are processed independently inside the system (no interaction) (superposition)

If we have several input signals $x_k[n]$ and their responses $y_k[n] = \mathcal{H}(x_k[n])$ then for the superposition of input signals $x[n] = \sum_k a_k x_k[n]$ we get the response $y[n] = \mathcal{H}(x[n])$ that is the superposition $y[n] = \sum_k a_k y_k[n]$.



We need:

- Some basic signals, $x_k[n]$, that can be used as basis for expanding a large class of $x[n]$ input signals.
- Easy way to calculate the response of LTI the basic signals.
- Easy way to determine the coefficients a_k , preferably independently of each other.

Signal decomposition into impulses

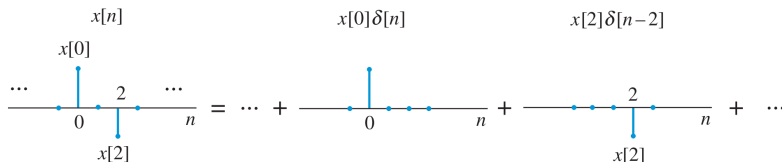
$$x_k[n] = \begin{cases} x[k], & n = k \\ 0, & n \neq k \end{cases}, \quad \delta[n - k] = \begin{cases} 1, & n = k \\ 0, & n \neq k \end{cases}$$

$$x[n] = \sum_{k=-\infty}^{\infty} x_k[n], \quad x_k[n] = x[k]\delta[n - k]$$

$$\rightarrow x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n - k].$$

Any signal can be decomposed into the superposition of scaled and delayed impulses.

Example



Example: the unit step

$$u[n] = \delta[n] + \delta[n - 1] + \delta[n - 2] + \dots = \sum_{k=0}^{\infty} \delta[n - k] = \sum_{k=-\infty}^n \delta[k].$$

We can determine the output of any LTI system if we know its impulse response:

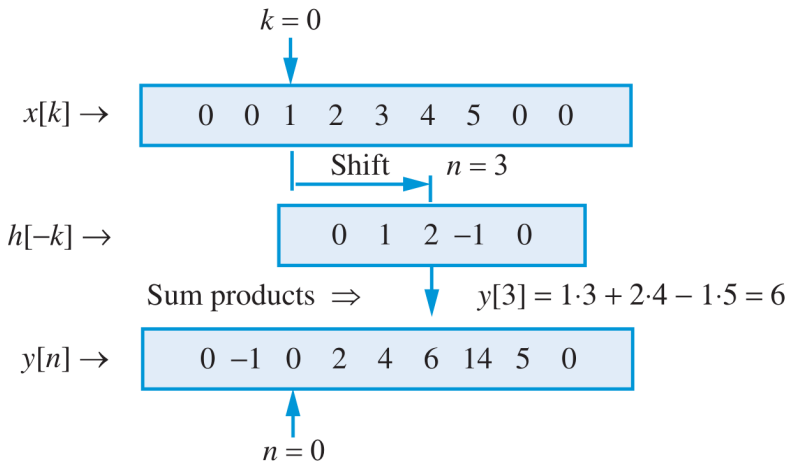
$$\left. \begin{array}{l} \delta[n] \xrightarrow{\mathcal{H}} h[n] \quad \text{impulse response} \\ \mathcal{H} \text{ time-invariant} \rightarrow \delta[n-k] \xrightarrow{\mathcal{H}} h[n-k] \\ x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] \quad |\mathcal{H}(\dots)| \end{array} \right\} \xRightarrow{\mathcal{H} \text{ linear}} x[n] \xrightarrow{\mathcal{H}} y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

Exercise

Test the time-invariance of the above convolution sum

Example

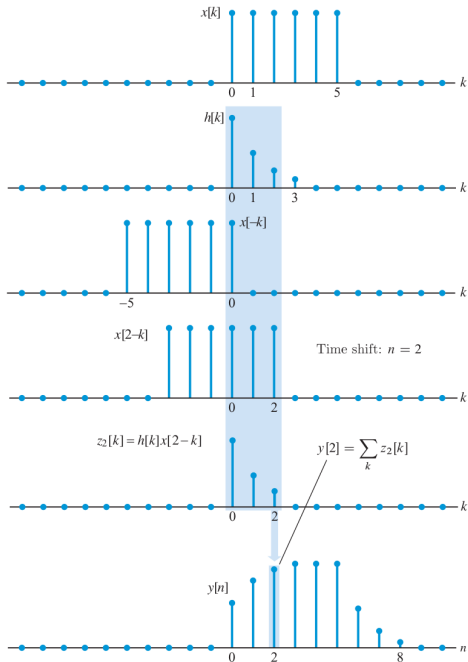
$$x[n] = \{1, 2, 3, 4, 5\}, \quad h[n] = \{-1, \mathbf{2}, 1\} \quad (\text{bold} \rightarrow n = 0)$$



Exercise

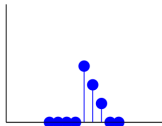
Write up the above convolution as an $\mathbf{y} = \mathbf{H} \cdot \mathbf{x}$ matrix multiplication where $\mathbf{y}$, $\mathbf{x}$ are column matrices and $\mathbf{H}$ is a rectangular matrix.

Example

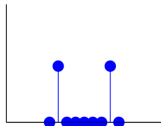


Example

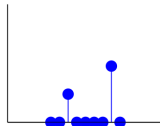
A



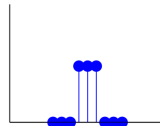
B



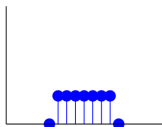
C



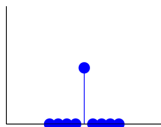
D



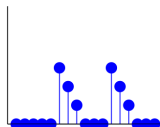
E



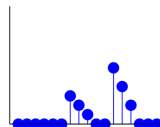
F



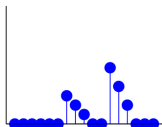
$A * B$



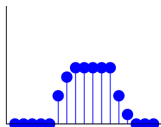
$A * C$



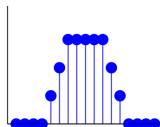
$C * A$



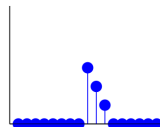
$A * E$



$D * E$



$A * F$



$$x[n] * h[n] \equiv \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

identity $x[n] * \delta[n] = x[n]$

delay $x[n] * \delta[n - n_0] = x[n - n_0]$

commutative $x[n] * h[n] = h[n] * x[n]$

associative $(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n])$

bilinear $x[n] * (a \cdot h_1[n] + b \cdot h_2[n]) = a \cdot x[n] * h_1[n] + b \cdot x[n] * h_2[n]$

Exercise

- 1 Prove the above properties of the convolution sum.
- 2 A finite-length sequence is non-zero only at a finite number of positions. If m and n are the first and last non-zero positions, respectively, then we call $n - m + 1$ the length of that sequence. What maximum length can the result of convolving two sequences of length k and l have?

Theorems

A linear time-invariant system with impulse response $h[n]$ is

- **causal** if

$$h[n] = 0 \quad \text{for} \quad n < 0 .$$

- **stable** in the bounded-input bounded-output (BIBO) sense, if and only if the impulse response is absolutely summable, that is, if

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty .$$

Proofs

- **causality**

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] , \quad h[n-k] = 0 \quad \text{for} \quad n-k < 0 \quad \text{i.e. for} \quad k > n$$

$$\Rightarrow y[n] = \sum_{k=-\infty}^n x[k]h[n-k] = \dots + x[n-2]h[2] + x[n-1]h[1] + x[n]h[0]$$

- **stability**

$$|x[n]| \leq M_x \quad \Rightarrow \quad |y[n]| \leq \sum_{k=-\infty}^{\infty} |x[k]| |h[n-k]| \leq M_x \sum_{k=-\infty}^{\infty} |h[n-k]| \quad \Rightarrow \quad \text{bounded}$$

- $n, k \rightarrow t, \tau$
- $y[n], x[k] \rightarrow y(t), x(\tau)$
- $\sum_k \rightarrow \int \cdots d\tau$
- $\delta[n] \rightarrow \delta(t)$ **Dirac-delta** continuous-time impulse
- $u[n] \rightarrow H(t)$ **Heaviside (step) function**

Convolution integral

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau .$$

1

$$\delta(x) = \begin{cases} 0 & ; \quad x \neq 0 \\ \infty & ; \quad x = 0 \end{cases}$$

2

$$\int_{-\infty}^{\infty} \delta(x) \, dx = 1,$$

Can produce such a "function" as the limiting case of a parameter dependent family of functions

1

$$\delta(n, x) = \begin{cases} n & |x| < \frac{1}{2n} \\ 0 & |x| > \frac{1}{2n} \end{cases}, \quad n \rightarrow \infty$$

2

$$\delta(\alpha, x) = \frac{1}{\pi} \frac{\alpha}{x^2 + \alpha^2}, \quad \alpha \rightarrow 0.$$

3

$$\delta(\alpha, x) = \frac{1}{\alpha\sqrt{\pi}} e^{-\frac{x^2}{\alpha^2}}, \quad \alpha \rightarrow 0.$$

4

$$\delta(\alpha, x) = \frac{\alpha/2}{\cosh^2 \alpha x}, \quad \alpha \rightarrow \infty$$

Properties

1

$$\int_{-\infty}^{\infty} f(x)\delta(x) \, dx = f(0);$$

Proof:

$$\int_{-\infty}^{\infty} f(x)\delta(x) \, dx \stackrel{\varepsilon \rightarrow 0}{\equiv} \int_{-\varepsilon}^{\varepsilon} f(x)\delta(x) \, dx = f(0) \int_{-\varepsilon}^{\varepsilon} \delta(x) \, dx = f(0).$$

2

$$\int_{-\infty}^{\infty} f(x)\delta(x - x_0) \, dx = f(x_0);$$

Proof:

$$x - x_0 = y, \quad \int_{-\infty}^{\infty} f(y + x_0)\delta(y) \, dy = f(x_0).$$

3

$$\delta(-x) = \delta(x);$$

4

$$\delta(ax) = \frac{\delta(x)}{|a|};$$

5

$$\delta(\phi(x)) = \sum_i \frac{\delta(x - x_i)}{|\phi'(x_i)|}, \quad \phi(x_i) = 0$$

Proof

$\delta(\phi(x))$ different from zero only around the roots of $\phi(x)$. In the close neighborhood of root x_i

$$\phi(x) \approx \phi(x_i) + (x - x_i)\phi'(x_i) = (x - x_i)\phi'(x_i):$$

$$\int_{-\infty}^{\infty} f(x)\delta(\phi(x))dx = \sum_i \lim_{\varepsilon \rightarrow 0} \int_{x_i-\varepsilon}^{x_i+\varepsilon} f(x)\delta(\phi(x))dx = \sum_i \lim_{\varepsilon \rightarrow 0} \int_{x_i-\varepsilon}^{x_i+\varepsilon} f(x)\delta((x - x_i)\phi'(x_i))dx = \dots$$

From properties 2. ($x_0 \equiv x_i$) and 4. ($a \equiv \phi'(x_i)$)

$$\dots = \sum_i \lim_{\varepsilon \rightarrow 0} \frac{1}{|\phi'(x_i)|} \int_{x_i-\varepsilon}^{x_i+\varepsilon} f(x)\delta(x - x_i)dx = \sum_i \frac{f(x_i)}{|\phi'(x_i)|}. \quad \square$$

6

$$\delta(x^2 - a^2) = \frac{\delta(x - a) + \delta(x + a)}{2|a|};$$

Examples:

$$\delta[(x - 1)(x - 2)(x - 3)] = \frac{\delta(x - 1)}{2} + \frac{\delta(x - 2)}{1} + \frac{\delta(x - 3)}{3};$$

$$\delta(\cos x) = \sum_{k=-\infty}^{\infty} \delta[x - (2k + 1)\frac{\pi}{2}];$$

7

$$\int_{-\infty}^{\infty} f(x) \delta'(x) \, dx = -f'(0);$$

Proof:

$$\int_{-\infty}^{\infty} f(x) \delta'(x) \, dx = f(x) \delta(x) \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f'(x) \delta(x) \, dx = -f'(0).$$

8

$$\int_{-\infty}^{\infty} f(x) \delta^{(n)}(x) \, dx = (-1)^n f^{(n)}(0);$$

Definition

$$H(x) = \begin{cases} 1 & , x > 0 \\ 0 & , x < 0 \end{cases}$$

Properties:

1

$$H'(x) = \delta(x)$$

2

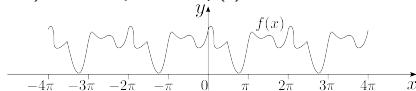
$$H'(x - x_0) = \delta(x - x_0)$$

Definition: periodic function

The period of function $f(x)$ is P if

$$f(x) = f(x \pm P) = \cdots = f(x \pm nP) = \cdots \quad n \in \{1, 2, 3, \dots\}.$$

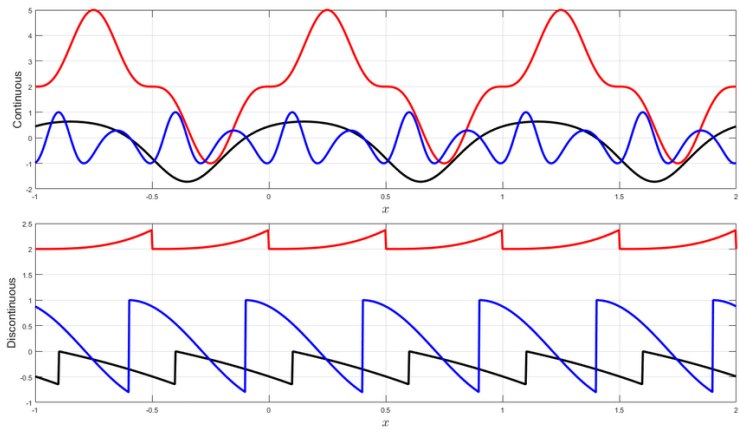
Similarly, if $\varphi(t) = \varphi(t + T)$ then the period of $\varphi(t)$ is T .



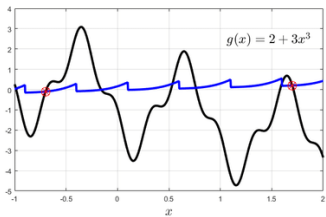
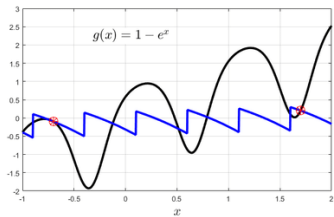
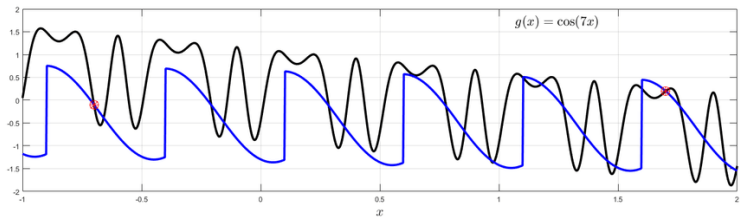
Most common periodic functions $\sin x$, $\cos x$ with period 2π .

$$\sin x = \sin(x + 2\pi), \quad \cos x = \cos(x + 2\pi)$$

$\sin 2x, \sin 3x, \dots, \sin nx, \dots$ and $\cos 2x, \cos 3x, \dots, \cos nx, \dots$ are also 2π periodic.
So are any function of these.



Periodic functions.



Periodic and non-periodic functions.

Properties

- $\alpha f_P(x) = h_P(x)$
- $\alpha + f_P(x) = h_P(x)$
- $f_P(x + \alpha) = h_P(x)$
- $f_P(x) + g_P(x) = h_P(x)$
- $f'_P(x) = h_P(x)$
- $f_P(x) \cdot g_P(x) = h_P(x)$
- $F(f_P(x)) = g_P(x)$
- $f_P(\alpha x) = f_{P/\alpha}(x)$
- $f_P(x) + g_{k \cdot P}(x) = h_{k \cdot P}(x)$
- $f_{P_1}(x) + g_{P_2}(x) = h_P(x)$, if $\exists n, m \in \mathbb{N}^*$, $P = nP_1 = mP_2$, i.e., $\frac{P_2}{P_1} = \frac{n}{m}$.

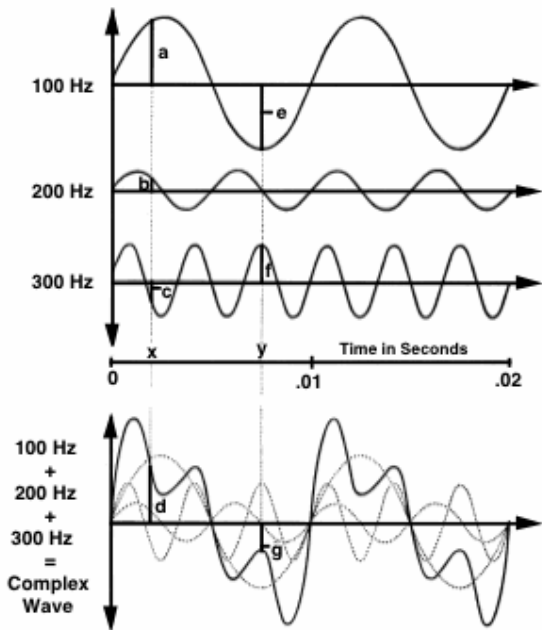
Fourier-theorem

For any 2π periodic function $f(x) = f(x + 2\pi)$ there exist the coefficients $a_0, a_1, \dots$ és $b_1, b_2, \dots$ so that

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) .$$

The r.h.s is a **trigonometric series**, the **Fourier series** of $f(x)$
 a_i and b_i are the function's **Fourier coefficients**.

- **Fundamental harmonic:** $a_1 \cos x + b_1 \sin x$
- **Higher harmonics:** $a_n \cos nx + b_n \sin nx$, $n \geq 2$



The integral over $[0, 2\pi]$ of the products of

$$\{1, \sin nx, \sin mx, \cos nx, \cos mx, \} \quad n \neq m, \quad n, m \in \{1, 2, 3, \dots\}$$

$$\int_0^{2\pi} \sin nx dx = 0, \quad n \in \mathbb{Z}, \quad \int_0^{2\pi} \cos nx dx = 0, \quad n \in \mathbb{Z}/\{0\},$$

$$\begin{aligned} & \int_0^{2\pi} \sin nx \sin mx dx = , \\ &= \int_0^{2\pi} \cos nx \cos mx dx = , \\ &= \int_0^{2\pi} \sin nx \cos mx dx = 0, \quad n \neq m, \end{aligned}$$

where we made use of

$$\begin{aligned} \sin nx \sin mx &= \frac{1}{2} [\cos(n - m)x - \cos(n + m)x], \\ \sin nx \cos mx &= \frac{1}{2} [\sin(n + m)x + \sin(n - m)x]. \end{aligned}$$

Scalar product of functions

For a given interval, e.g. $[0, 2\pi]$:

$$(\phi, \psi) \equiv \int_0^{2\pi} \phi(x)\psi(x) \, dx$$

If $(\phi, \psi) = 0 \rightarrow$ *orthogonal functions*.

Norm of a function

(ϕ, ϕ) a *norm squared of the function*.

$$\begin{aligned} (\phi, \phi) &\equiv \|\phi\|^2 \\ \|\phi\| &\geq 0, \quad \|\phi\| = 0 \iff \phi = 0. \end{aligned}$$

Scalar product of the trigonometric basis functions



$$(1, \sin nx) = (1, \cos nx) = 0, \quad (\sin nx, \sin mx) = (\cos nx, \cos mx) = (\sin nx, \cos mx) = 0$$

→ orthogonal.



$$\|1\|^2 = \int_0^{2\pi} 1^2 dx = 2\pi,$$



$$\|\sin nx\|^2 = \int_0^{2\pi} \sin^2 nx dx = \pi,$$



$$\|\cos nx\|^2 = \int_0^{2\pi} \cos^2 nx dx = \pi.$$



$$(1, \sin nx) = (1, \cos nx) = (\sin nx, \cos mx) = 0 \quad \forall \quad n, m \in \mathbb{Z}$$

$$(\sin nx, \sin mx) = (\cos nx, \cos mx) = \pi \delta_{nm} = \begin{cases} \pi & n = m \\ 0 & n \neq m \end{cases} \quad \{n, m\} \in \mathbb{N}$$

δ_{nm} -Kronecker-delta.

Then we can determine the Fourier coefficients of the expansion based on the equality

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Integrating both sides over $[0 - 2\pi]$:

$$\int_0^{2\pi} f(x) \, dx = a_0 \pi \quad \Rightarrow \quad a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) \, dx$$

Multiplying both sides by $\cos mx$ (and $\sin mx$) and integrating from 0 to 2π :

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx \, dx, \quad n = \{0, 1, 2, \dots\}$$
$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx, \quad n = \{1, 2, \dots\}$$

For a 2π periodic function instead of the $0, 2\pi$ interval we can as well use the $-\pi, +\pi$ or for that matter any $\alpha, 2\pi - \alpha$ interval.

Example

Let $f : [-\pi, \pi] \rightarrow \mathbb{R}$, 2π periodic function:

$$f(x) = \begin{cases} 0, & x \in [-\pi, 0] \\ x, & x \in [0, \pi) \end{cases}$$



$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} x dx = \frac{\pi}{2},$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_0^{\pi} x \cos nx dx =,$$

$$= \frac{1}{\pi} x \frac{\sin nx}{n} \Big|_0^{\pi} - \frac{1}{\pi} \int_0^{\pi} \frac{\sin nx}{n} dx = \frac{1}{\pi} \frac{\cos nx}{n^2} \Big|_0^{\pi} = \frac{1}{\pi} \frac{(-1)^n}{n^2} - \frac{1}{\pi} \frac{1}{n^2} = \begin{cases} 0, & \text{if } n \text{ even} \\ -\frac{2}{\pi n^2}, & \text{if } n \text{ odd} \end{cases}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_0^{\pi} x \sin nx dx = -\frac{1}{\pi} x \frac{\cos nx}{n} \Big|_0^{\pi} + \frac{1}{\pi} \int_0^{\pi} \frac{\cos nx}{n} dx =$$

$$= -\frac{(-1)^n}{n} + \frac{1}{\pi} \frac{\sin nx}{n^2} \Big|_0^{\pi} = \frac{(-1)^{n+1}}{n}.$$

$$f(x) = \frac{\pi}{4} - \frac{2}{\pi} \left(\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \cdots \right) + \frac{\sin x}{1} - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} + \cdots,$$

Moving from $[0, 2\pi)$ to $[-\pi, \pi)$ we can exploit the functions symmetry properties.

1

$$f(x) \text{ even} \Rightarrow f(x) = f(-x)$$

As $\sin nx = -\sin(-nx)$ odd (antisymmetric),

$$b_n = 0, \quad n \in \mathbb{N} \quad \text{and}$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \, dx \Rightarrow$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) \, dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx \Rightarrow$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx \quad n \in \mathbb{N}$$

2

$$f(x) \text{ odd} \Rightarrow f(x) = -f(-x)$$

As $\cos nx = \cos(-nx)$ even (symmetric),

$$a_n = 0 \quad n \in \{0, 1, 2, \dots\} \quad \text{and}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx \Rightarrow$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx, \quad n \in \mathbb{N}$$

Example: rectangular wave (pulse train)

Let

$$f(x) = \begin{cases} -1 & x \in [-\pi, 0) \\ 1 & x \in [0, \pi) \end{cases}$$

odd function, i.e. $a_0 = a_n = 0$ and

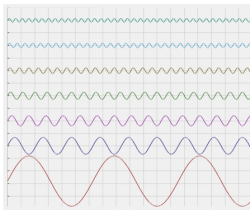
$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} \sin nx \, dx = - \left. \frac{2 \cos nx}{\pi n} \right|_0^{\pi} = \frac{2(-1)^{n+1}}{\pi n} + \frac{2}{\pi n}$$

or

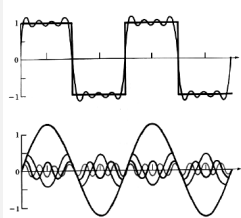
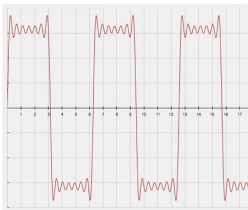
$$b_n = \begin{cases} \frac{4}{\pi n} & , n \text{ even} \\ 0 & , n \text{ odd} \end{cases}$$

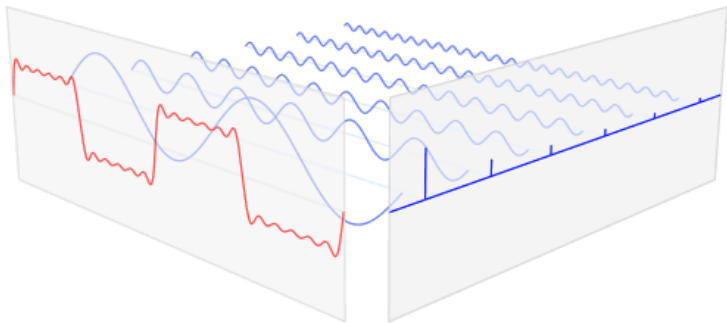
Therefore

$$f(x) = \frac{4}{\pi} \left(\frac{\sin x}{1} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \dots \right)$$



=





Example: triangle wave (pulse train)

$$f(x) = |x| \quad x \in (-\pi, \pi), \quad f(x) = f(-x) \Rightarrow b_n = 0 \quad n \in \mathbb{N}$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) \, dx = \frac{2}{\pi} \int_0^{\pi} x \, dx = \pi$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx = \frac{2}{\pi} \int_0^{\pi} x \cos nx \, dx$$

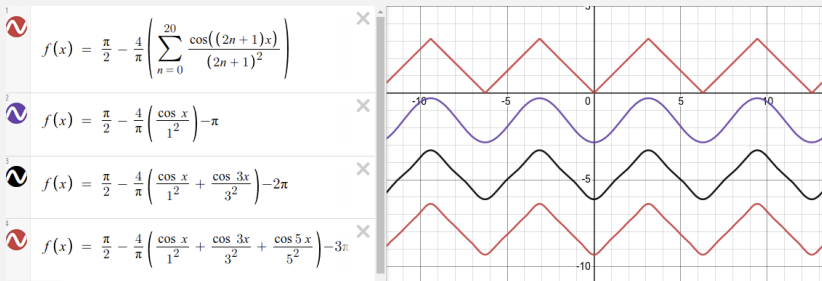
integrated by parts

$$a_n = \frac{2}{\pi} x \frac{\sin nx}{n} \Big|_0^{\pi} - \frac{2}{\pi} \int_0^{\pi} \frac{\sin nx}{n} \, dx = \frac{2 \cos nx}{\pi n^2} \Big|_0^{\pi} = \frac{2(-1)^n}{\pi n^2} - \frac{2}{\pi n^2}$$

$$a_n = \begin{cases} 0, & n \text{ even} \\ -\frac{4}{\pi n^2}, & n \text{ odd} \end{cases}$$

therefore

$$f(x) = \frac{\pi}{2} - \frac{4}{\pi} \left(\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right)$$



Example: sawtooth wave

Expand the periodic function

$$f(x) = \frac{\pi - x}{2}$$

over the $(0, 2\pi)$ interval.

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} \frac{\pi - x}{2} dx = \frac{1}{2\pi} \left(\pi x - \frac{1}{2} x^2 \right) \Big|_0^{2\pi} = 0,$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} \frac{\pi - x}{2} \cos nx dx = \frac{1}{2\pi} (\pi - x) \frac{\sin nx}{n} \Big|_0^{2\pi} - \frac{1}{2n\pi} \int_0^{2\pi} \sin nx dx = 0,$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} \frac{\pi - x}{2} \sin nx dx = - \frac{1}{2\pi} (\pi - x) \frac{\cos nx}{n} \Big|_0^{2\pi} - \frac{1}{2n\pi} \int_0^{2\pi} \cos nx dx = \frac{1}{n}.$$

Therefore

$$\frac{\pi - x}{2} = \sum_{n=1}^{\infty} \frac{\sin nx}{n} \quad (0 < x < 2\pi). \quad (1)$$



$$f(x) = \sum_{n=1}^{60} \frac{\sin(nx)}{n}$$



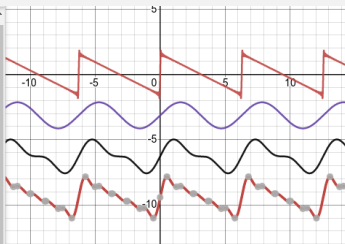
$$f(x) = \sin(x) - \pi$$



$$f(x) = \sin(x) + \frac{\sin(2x)}{2} - 2\pi$$



$$f(x) = \sin(x) + \frac{\sin(2x)}{2} + \frac{\sin(3x)}{3} + \frac{\sin(4x)}{4} + \frac{\sin(5x)}{5} - 3\pi$$



Example

Find the Fourier series of the symmetric $f(x) = x^2$ function over the $(-\pi, \pi)$ interval.

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x^2 dx = \frac{\pi^2}{3},$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx dx = \dots = (-1)^n \frac{4}{n^2} \quad n \in \mathbb{N},$$

$$x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} (-1)^n \frac{\cos nx}{n^2} \quad (-\pi \leq x \leq \pi).$$

